

Navigating The Future of VFX Education

Rico Dober
Filmuniversität Babelsberg KONRAD
WOLF
Potsdam, Germany
r.dober@filmuniversitaet.de

Josh Planz
School of Visual Arts
New York, USA
jplanz@sva.edu

Shaun Roth
Industrial Light & Magic
Vancouver, Canada
shaun.roth@gmail.com

Andreas Maaninka
Oasis VFX
Vancouver, Canada
info@oasisvfx.com

Agon Ushaku
Polygrail GmbH
Stuttgart, Germany
agon.ushaku@polygrail.de



Figure 1: Students setting up a virtual production film set at LEDcave Berlin.

Abstract

This panel examines the relevance of traditional VFX techniques in the rapidly evolving industry, discussing how education can integrate these foundational skills with new technologies like AI and real-time rendering. Featuring industry leaders and educators, the session seeks strategies to adapt curricula to prepare students for modern VFX demands.

CCS Concepts

• **Human-centered computing** → Mixed / augmented reality; Virtual reality; • **Computing methodologies** → Artificial intelligence; Machine learning; Computer graphics; Animation; Rendering; Image manipulation; Shape modeling; • **Applied computing** → Media arts; Performing arts; Fine arts; • **Social and professional topics** → Model curricula; **Computer science education**; Intellectual property.

Keywords

VFX Education, Visual Effects, Virtual Production, Realtime Rendering, Artificial Intelligence

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1 Introduction

The VFX industry is currently navigating a complex landscape characterized by rapid technological advancements, increased demand for high-quality content, and evolving global workflows. These challenges require a reevaluation of traditional education models to ensure that future professionals are well prepared for industry realities. Historically, traditional VFX techniques have provided a foundational framework that underscores the principles of visual effects, yet the emergence of real-time rendering, virtual production, and artificial intelligence poses questions about the ongoing relevance of these legacy methods.

In this panel, featuring prominent educators and industry leaders, we will explore whether traditional VFX skills continue to hold value or require modernization to meet contemporary expectations. The discussion will focus on how educational institutions can adapt by potentially integrating these time-honored techniques with cutting-edge technologies. The aim is to prepare students for the demands of a globalized industry environment, where efficient workflows, ethical considerations, and sustainable practices are gaining importance.

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Panelists will debate the balance necessary between preserving the foundational skills that support creative and technical excellence and the need to embrace innovative methodologies that respond to industry shifts. The conversation will also consider how education can instill adaptability, critical thinking, and the ability to seamlessly integrate both traditional and new VFX techniques. By critically assessing these elements, the panel aims to provide actionable insights into curriculum development that aligns with industry trends, ensuring students are equipped to thrive in a competitive and evolving field. Through this dialogue, the session seeks to forge a path forward for VFX education that honors traditional techniques while fostering innovation and adaptability. Our panel consists of:

- **Rico Dober (Moderator)** is a professor of Visual Effects at Film University Babelsberg KONRAD WOLF and Dean of Studies in the Bachelor's program Visual Effects & Virtual Productions. After studying Media and Information Sciences, he worked as a freelance Digital Artist and Technical Director in Berlin. In 2009 he moved to London, joining Framestore as Lead Pipeline TD on "Gravity". At DNEG, he held roles like FX Supervisor and CG Supervisor, contributing to major Hollywood productions. In 2020, he returned to Germany as a professor at TH Ostwestfalen-Lippe. He is a board member of the Visual Effects Society Germany.
- **Andreas Maaninka**, originally from Stockholm, is the founder and independent Visual Effects Supervisor at Oasis VFX. He began his career in the early 2000s by moving to London, working at Pinewood Studios and then in Soho. After spending time in New Zealand and Tokyo, he has settled in Vancouver for the past 14 years. He has worked at prestigious companies like Weta, Framestore, Sony Pictures Imageworks, and Digital Domain, contributing to films such as "Bladerunner 2049," "Avatar," and "Batman The Dark Knight." Currently, he focuses on the production side as an on-set VFX supervisor, directly collaborating with filmmakers to shape the story and vision. Education and continuous learning are important to him, and he prioritizes elevating himself and his team, viewing it as ongoing mentorship. Additionally, he participates in global speaking engagements, addressing various subjects in the spirit of education.
- **Joshua Planz** is a Brooklyn-based 3D artist and filmmaker with a versatile body of work spanning film, television, and independent media. With over a decade of experience in visual effects and animation, his portfolio ranges from emotionally driven short films to vibrant, imaginative content for young audiences. He has collaborated with top studios including The Mill, Psyop, Hornet, and FuseFX, and currently serves as Head of 3D at HOPR Studios. Recent credits include Barbie and American Horror Story. In addition to his professional practice, Josh has been teaching 3D animation at the School of Visual Arts for over ten years, mentoring students at nearly every stage of the production pipeline, from foundational first-year courses to advanced thesis development.
- **Shaun Roth** is a CG Supervisor with 17 years of experience in visual effects. He recently worked on "Percy Jackson and

the Olympians" as part of ILM's StageCraft crew and previously supervised "Willow", "Y: The Last Man", and "The Witcher" (Season 2). He also contributed to "A Quiet Place: Day One" and "Skeleton Crew". In 2022, he received an Engineering Emmy for his work on ILM's StageCraft virtual production technology. Before joining ILM, Roth worked in the field of FX on such films as the Oscar-winning film "Interstellar" by Christopher Nolan, working on the look and development of the Gargantua Black Hole. Roth played a key role in launching DNEG Vancouver and later helped establish DNEG Montreal.

- **Agon Ushaku** has over 20 years of experience in the 3D/CGI industry, beginning with a notable 1998 animation for Novartis Pharma. Since then, he has worked on feature films, commercials, and industrial animations for major clients like Bosch, Daimler, Samsung, and Vodafone. Beyond his technical expertise, he is a consultant in visual solutions for global clients. He serves as Chair of the VES Germany Section and Curator of the VES Megabrain Masterclasses, promoting industry collaboration. Additionally, he is a respected educator, teaching 2D/3D Design and 3D Modeling at Macromedia University of Applied Sciences in Stuttgart.

This work complements existing educational strategies by offering an integrative model that respects traditional VFX roots while embracing necessary technological shifts. It provides a critical forum for reimagining how VFX education can evolve to better satisfy industry demands and graduate expectations, contributing to SIGGRAPH's mission of fostering innovation and excellence in computer graphics and interactive techniques.

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